

DSA / Algorithms (Abdul Bari) - Day 42

45-day plan mapped to the Algorithms playlist

NP-Hard & NP-Complete

Complexity classes P, NP, NP-Hard, NP-Complete.

Watch these Abdul Bari videos

- > 8. NP-Hard and NP-Complete Problems
- > 8.1 NP-Hard Graph Problem - Clique Decision Problem

Key concepts

- P: solvable in polynomial time; NP: verifiable in poly time.
- NP-Complete = hardest in NP; NP-Hard $\geq$ NP.
- Reductions show hardness.

Key idea

Clique decision is NP-Complete.

Tasks (1.5h)

- ☐ Watch both videos
- ☐ Define P/NP/NPC/NPH
- ☐ Explain a reduction

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0